

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026

Biology — Class IX | Paper: SSC-I

Syllabus: Ch 1 — The Science of Biology | Ch 2 — Biodiversity | Ch 3 — The Cell | Ch 4 — Cell Cycle
| Ch 5 — Tissues, Organs & Organ Systems | Ch 6 — Molecular Biology | Ch 7 — Metabolism (Sec
7.1-7.6)

ANSWER KEY (Version 3)

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

Q1. Study of drugs such as aspirin, morphine, insulin, and penicillin is called: **D — Pharmacology** **1 mark**

Pharmacology is the science that deals with the study of drugs. A drug is a chemical substance used to treat pain, fever, or inflammation — examples include aspirin, morphine, insulin, and penicillin. Pathology is the study/diagnosis of disease; Immunology is the study of immunity; Physiology is the study of organ functions.

Q2. A mule is not a true species because it is: **B — Infertile due to an odd number of chromosomes** **1 mark**

A cross between a male donkey and a female horse produces a **mule**, which is infertile because of the **odd number of chromosomes** it inherits — the two parents cannot reproduce successfully. A species is a group of similar organisms capable of interbreeding and producing **fertile offspring**, so the mule fails this test.

Q3. Described 350 species of animals in the 700s, before Linnaeus: **A — Abu Usama Aljahiz** **1 mark**

Abu Usama Aljahiz, in the 700s, described 350 species of animals — well before Linnaeus (1707-1778) introduced binomial nomenclature. Aristotle was the first person to classify living organisms (much earlier); Caesalpino (1519-1603) divided plants into fifteen genera; John Ray (1627-1705) published works on plant classification.

Q4. Lysosomes are "suicidal bags" because they:

D — Digest the cell's own unwanted organelles through autophagy **1 mark**

Lysosomes are single membrane-bound sacs containing digestive enzymes. Besides intracellular digestion of materials taken up from outside, they also engulf and digest the cell's own unwanted or worn-out organelles — this process is called **autophagy**. They do not produce ATP (mitochondria do this) or synthesize proteins (ribosomes do this).

Q5. During the G1 phase, the cell mainly:

D — Grows, increases organelles, and forms proteins to prepare for DNA synthesis **1 mark**

During **G1 (Gap 1) phase**, the cell grows in size, increases the number of its organelles, and forms the enzymes and nucleotides needed for the next phase (S phase, DNA replication). DNA is actually copied during the S phase, not G1; the spindle-forming proteins are mainly produced during G2.

Q6. NOT one of the four phases of karyokinesis in mitosis: **B — Synapsis** **1 mark**

Karyokinesis in **mitosis** is divided into four phases: Prophase, Metaphase, Anaphase, and Telophase. **Synapsis** is the point-to-point pairing of homologous chromosomes that occurs only during **Prophase I of meiosis** — it does not occur in mitosis, since mitosis does not involve homologous chromosome pairing.

Q7. Organ system with pituitary, thyroid, parathyroid, pancreas, adrenal glands: **A — Endocrine system**

1 mark

The **endocrine system** provides communication within the body via hormones and directs long-term change in other organ systems. Its organs/structures include the pituitary, pineal, thyroid, parathyroid, pancreas, adrenals, testes and ovaries. It maintains balance of blood components like glucose, water, and calcium.

Q8. Test that detects protein by turning pink/violet: **C — Biuret test** **1 mark**

In the **Biuret test**, a protein solution is treated with NaOH and copper sulphate solution; a **pink or violet colour** confirms the presence of protein. The iodine test detects starch (blue-black), Benedict's test detects reducing sugars (colour change on heating), and the emulsion test detects lipids (cloudy white suspension).

Q9. Cholesterol belongs to the lipid class with a three-ring structure: **C — Steroids** **1 mark**

Steroids do not have fatty acids in their formula; instead they have a characteristic three-ring structure, e.g. cholesterol and vitamin D. Cholesterol, along with phospholipids, is a component of the plasma membrane structure. Fatty acids and phospholipids are different classes of lipids that do contain fatty acid chains; waxes are used by insects to build hives.

Q10. Amylase acting only on starch, not proteins/fats, shows enzyme: **B — Specificity** **1 mark**

Enzymes are usually very **specific**: each enzyme catalyses only one type of reaction and will not act on a different substrate. For example, **amylase** will only act on starch and not on proteins or fats. This is different from cofactor requirement (a non-protein helper molecule) or regulation (control of enzyme synthesis/activity).

Q11. Pepsin working outside the cells that produced it is a(n): **C – Extracellular enzyme** **1 mark**

Enzymes that are made inside cells but released to work outside are called **extracellular enzymes**, e.g. **pepsin** working in the stomach cavity. Intracellular enzymes stay and work inside the cell. A coenzyme is a detachable organic cofactor; a ribozyme is an RNA molecule that acts as an enzyme.

Q12. Without enzymes, a biochemical reaction may take: **A – Months or years to complete** **1 mark**

A reaction may initiate without the involvement of enzymes, but it will be **extremely slow** and may take **months or years** to complete. Enzymes speed up such reactions **millions of times** so that they can occur at body temperature within seconds. Enzymes only speed up a reaction — they do not start it.

SECTION B — Short Questions (11 × 3 = 33 Marks)

2(ii) State any three Quranic references relating to the origin and development of life, with Sura names.

3 marks

1. **Origin of life in water** — Sura Al-Anbiya (Ambia) 21, Ayat 30: "We made every living thing from water." As living things consist of 60-90% water, all living things have a common origin. **1 mark**
2. **Creation of man** — Sura Ar-Rahman, Ayat 14: "He made man from clay like the potter." **1 mark**
3. **Development** — Sura Al-Mominoon, Ayat 14: describes the sequence of developmental stages of the embryo (a clot, then a little lump, then bones, then flesh). **1 mark**

OR

Name any three Muslim scientists mentioned in Chapter 1, and state the three major fields of biology.

3 marks

Muslim scientists (any three): **Jabir Bin Hayyan**, **Abdul Malik Asmai**, and **Bu Ali Sina** — their knowledge contributed greatly to the development of present-day knowledge of plants and animals. **1.5 marks**

Three major fields of biology: **Botany** (study of plants), **Zoology** (study of animals), and **Microbiology** (study of micro-organisms, e.g. viruses, bacteria). **1.5 marks**

2(ii) State the four aims of classification mentioned in the textbook. **3 marks**

1. To **determine similarities and differences** between organisms.
2. To **arrange organisms** on the basis of similarities and differences.
3. To **identify the organisms** to study them systematically.
4. To **find out evolutionary relationships** among organisms.

0.75 marks each

OR

Differentiate between autotrophic, heterotrophic, and saprotrophic modes of nutrition. **3 marks**

Autotrophs are organisms capable of producing their own food (photosynthetic mode of nutrition), e.g. green plants. **1 mark**

Heterotrophs are organisms which eat other things as food (ingestive mode of nutrition), e.g. animals. **1 mark**

Saprotrophs are organisms that depend on dead, decaying matter (absorptive mode of nutrition), e.g. fungi and bacteria. **1 mark**

2(iii) Differentiate between prions and viroids as acellular infectious agents. 3 marks

Prions are composed of **proteins only**. 1 mark

Viroids are composed of **circular RNA only**. 1 mark

Both cause infectious diseases in certain plants, both are **acellular particles**, and neither is included in any domain or kingdom of the modern classification system. 1 mark

OR

Write the taxonomic hierarchy (Kingdom to Species) for the classification of a Pea plant. 3 marks

Taxon	Pea
Kingdom	Plantae
Division	Magnoliophyta
Class	Magniopsida
Order	Fabales
Family	Fabaceae
Genus	<i>Pisum</i>
Species	<i>Pisum sativum</i>

~0.43 marks per correct rank (any 7)

2(iv) Describe the three layers of a plant cell wall and the function of plasmodesmata. 3 marks

Middle lamella: made up of magnesium and calcium salts of pectin; sticky in nature; holds the neighbouring cell walls together. 1 mark

Primary wall: contains cellulose fibres arranged in a criss-cross fashion; thin and flexible. 1 mark

Secondary wall: formed inside the primary wall in some cells (e.g. xylem vessels); very thick and rigid due to the presence of lignin. 0.5 marks

Plasmodesmata: tiny pores in the cell wall through which neighbouring cells form cytoplasmic connections. 0.5 marks

OR

Describe the structure of the plant vacuole, the tonoplast, and turgor pressure. 3 marks

A **vacuole** is a membrane-bound, fluid-filled sac. Plant cells have one **large central vacuole** formed by the joining of small vacuoles. 1 mark

The membrane of the plant vacuole is called the **tonoplast**. It contains a liquid called **cell sap**, which has dissolved materials like mineral salts, sugars, and amino acids. 1 mark

Turgor pressure occurs when the fluid content of the cell pushes the cell membrane against the cell wall, providing shape and support to the plant cell. 1 mark

2(v) Describe any three specialized cell types and how structure suits function. 3 marks

Red blood cells (RBCs): biconcave disk shape gives more surface area to absorb/release oxygen; lack nucleus, mitochondria and ER to accommodate more haemoglobin; flexible enough to pass through capillaries; average age is 120 days. 1 mark

Neurons: have a cell body plus two types of cytoplasmic fibres — **dendrites** (conduct impulses to the cell body) and **axons** (conduct messages away from the cell body) — enabling coordination across far-away cells of the body. 1 mark

Muscle cells: elongated in shape and filled with **actin** and its associated contractile proteins, giving them the ability to contract and relax (three types: smooth, striated, cardiac). 1 mark

OR

Describe the structure and function of ribosomes; how do prokaryotic and eukaryotic ribosomes differ? 3 marks

Ribosomes are composed of roughly equal amounts of **proteins and ribosomal RNA (rRNA)**, made up of two subunits — one large, one small — which join together to perform their function. 1 mark

They are tiny granular structures, **not bound by any membrane**, found in both prokaryotic and eukaryotic cells; proteins make up about 55% of the dry weight of a cell, much of which is synthesized by ribosomes. 1 mark

Prokaryotic ribosomes are smaller in size than eukaryotic ribosomes. 1 mark

2(vi) Down syndrome (47 chromosomes, extra chromosome 21) — what goes wrong during meiosis.

3 marks

Normally, during **meiosis I**, each pair of homologous chromosomes separates so that every gamete receives only **one member of the pair** — giving a haploid gamete with 23 chromosomes in humans. **1 mark**

If a pair of homologous chromosomes (or a pair of sister chromatids during meiosis II) **fails to separate properly**, one resulting gamete ends up with an **extra chromosome** (24 instead of 23), while the other gamete has one too few. **1 mark**

When the gamete carrying the extra chromosome 21 fuses with a normal gamete at **fertilization**, the zygote formed has **47 chromosomes** instead of 46 — producing Down syndrome. **1 mark**

OR

Define chromatid and centromere. Explain how sister chromatids are formed during the S phase of interphase. **3 marks**

Chromatid: one of the two identical copies of a chromosome, formed after DNA replication. **1 mark**

Centromere: the point at which two sister chromatids are attached to each other. **1 mark**

During the **S (Synthesis) phase** of interphase, the cell replicates its entire genetic material to form two exact copies; that is why each chromosome now has two **sister chromatids**, one for each future daughter cell. **1 mark**

2(vii) Organs of the lymphatic and immune systems, and one homeostatic role each. **3 marks**

Lymphatic system: organs/structures are lymph, lymph nodes, and lymph vessels; role — defends against infection/disease and **maintains tissue fluid homeostasis**. **1.5 marks**

Immune system: organs/structures are leukocytes, tonsils, adenoids, thymus, and spleen; role — **removes pathogens**, fights infections, and helps in healing. **1.5 marks**

OR

How do the muscular and integumentary systems help maintain body temperature? **3 marks**

Muscular system (skeletal, cardiac, smooth muscles): provides movement and support, and **helps in temperature regulation by heat production** (e.g. during exercise, muscle activity generates heat). **1.5 marks**

Integumentary system (skin, hair, nails): provides protection from injury and fluid loss, and is directly **involved in temperature regulation** (e.g. sweating, hair response). **1.5 marks**

2(viii) Two textbook examples of restoring homeostasis (exercise; fruit juice). **3 marks**

During **exercise**, the muscles increase heat production, pushing body temperature upward; the body opposes this by **increasing blood flow to the skin, sweating, and heavy breathing** to cool down and restore the balance point. **1.5 marks**

After drinking a glass of **fruit juice**, blood glucose concentration goes up; homeostasis is the body's ability to detect this change and work to bring glucose **back down to its balance point**. **1.5 marks**

OR

Organs of the cardiovascular and skeletal systems, and their homeostatic role. **3 marks**

Cardiovascular system: heart, blood, and blood vessels; helps **stabilize levels of gases, wastes, body temperature and pH**. **1.5 marks**

Skeletal system: bones, cartilage, joints, tendons, and ligaments; regulates the level of **calcium and other minerals** in the blood by storing or releasing them from bones. **1.5 marks**

2(ix) Organs of the nervous and respiratory systems, and their homeostatic role. 3 marks

Nervous system: brain, spinal cord, nerves, and sensory organs (eyes, ears, tongue, skin, nose); maintains homeostasis by **rapidly controlling body functions** and directing short-term change in other systems.

1.5 marks

Respiratory system: mouth, nose, pharynx, larynx, trachea, bronchi, lungs, and diaphragm; controls the balance of **oxygen and carbon dioxide** in the body. **1.5 marks**

OR

Define homeostasis; name two variables (other than temperature) the body keeps stable. 3 marks

Homeostasis is the tendency to maintain a stable, relatively constant internal environment. **1 mark**

Variable 1: concentration of various **ions** — e.g. the stomach maintains a pH different from that of surrounding organs, and each cell maintains its own ion concentrations. **1 mark**

Variable 2: concentration of **glucose** — e.g. blood glucose is kept within a balanced range after eating or exercising. **1 mark**

2(x) Describe the structure of an amino acid (four groups on the central carbon). 3 marks

Each amino acid contains a central carbon atom called the **α (alpha) carbon**. **0.5 marks**

Four different groups are attached to the α-carbon: a **hydrogen atom (-H)**, **0.5 marks**

an **amino group (-NH₂)**, **0.75 marks**

a **carboxyl group (-COOH)**, and an **alkyl (-R) group**. Each amino acid has a different R group, which determines its chemical properties (there are about 20–25 different types of amino acids). **1.25 marks**

OR

List the ten minor (trace) bioelements making up ~1% of protoplasm. 3 marks

Potassium, Sulphur, Chlorine, Sodium, Magnesium, Iron, Copper, Manganese, Zinc, and Iodine are the ten minor bioelements; together they make up only about 1% of the protoplasm of living organisms.

~0.3 marks each (any correct combination totalling 3 marks)

2(xi) What is a metabolic pathway? Role of enzymes in forming it. 3 marks

Many **enzymes work together in a specific sequence** to make metabolic pathways. **1 mark**

A **metabolic pathway** is a series of connected chemical reactions. **1 mark**

Multiple **intermediate molecules** are produced before the conversion of an initial substrate into the final product. **1 mark**

OR

Define catabolism and anabolism with an example of each (other than photosynthesis/respiration). 3 marks

Catabolism: reactions which break down complex molecules into simpler molecules. Example: **digestion** — a large, complex food molecule is broken down into smaller and simpler molecules. **1 mark**

Anabolism: reactions which build complex molecules from simpler molecules. Example: **condensation of glucose molecules to form glycogen** in muscle and liver cells. **1 mark**

Usually, **energy is released in catabolism** and it is **utilized in anabolism**. **1 mark**

SECTION C — Long Questions (4 × 5 = 20 Marks)

3(i) Name FIVE sub-fields of biology (other than Anatomy, Pharmacology) with one example/application each. 5 marks

Histology: the microscopic study of tissues of organisms, e.g. epithelial tissue that forms a continuous layer covering the entire body surface. **1 mark**

Genetics: the study of genes and heredity in organisms, e.g. plants having red flowers produce red-flowered offspring, while white cats produce white kittens. **1 mark**

Embryology: the study of the development of an organism from a fertilized egg. **1 mark**

Paleontology: the study of the history of life on Earth as based on fossils. **1 mark**

Immunology: the study of immunity — the ability of the body to protect itself from foreign substances/microbes. Example: immunization through vaccination protects against diseases like polio, corona, and dengue. **1 mark**

(Other acceptable sub-fields: Morphology, Cytology, Molecular biology, Taxonomy, Ecology, Marine biology, Pathology — any five with correct example score full marks.)

OR

Describe the relationship of biology with other sciences (Biophysics, Biochemistry, Biostatistics, Computational Biology, Biogeography). 5 marks

Biophysics: biological organisms work on the principles of physics, e.g. movement of muscles and bones; the study of biological phenomena according to the principles and laws of physics. **1 mark**

Biochemistry: the study of chemical constituents found in an organism and the chemical reactions taking place in the living organism, e.g. digestion of food, respiration, and photosynthesis. **1 mark**

Biostatistics: the collection of biological data/facts through observations and experiments, and analysing them according to statistical rules for biological study; also called biometry. **1 mark**

Computational biology: the study of the use of data analysis, mathematical modelling, and computational simulations to understand a biological system, e.g. locating fragments of DNA on chromosomes. **1 mark**

Biogeography: the study of the distribution of plants and animals in different geographical regions of the world. **1 mark**

3(ii) Structure and functions of centrioles and the three cytoskeletal fibres. 5 marks

Centrioles: hollow, open-ended, cylinder-like structures found only in animal cells; they exist in pairs near the nuclear envelope. Each centriole consists of nine triplets of microtubules. At the start of cell division, centrioles duplicate and two pairs migrate to opposite poles, helping form the spindle apparatus; centrioles are also involved in the formation of cilia and flagella. **2 marks**

Microtubules: made up of tubulin protein; unbranched, hollow, tube-like structures; give rise to spindle fibres, cilia, and flagella. **1 mark**

Microfilaments: very thin protein fibres consisting mainly of the contractile protein actin; responsible for the streaming movements of the cytoplasm; also regulate overall cell movement. **1 mark**

Intermediate filaments: composed of a variety of proteins including keratin and vimentin; form a branching network; maintain cell structure and, in tissues, fix cells to each other. **1 mark**

OR

Compare the structural advantages of plant and animal cells (any five), and describe stem cells with reference to haematopoietic stem cells. 5 marks

Plant Cell Advantage	Animal Cell Advantage
Cell wall cements adjoining cells, forming supportive structure.	Support depends on collective arrangement of tissues, organs, and skeletal systems (no cell wall needed).
Transport channels — xylem and phloem — are formed because of cell walls.	Transport of fluids, nutrients, and gases occurs through different structures and mechanisms.
Rigid cell wall helps withstand high osmotic stress and store water.	Cannot withstand high osmotic pressure or store large volumes of water.
Can become turgid, which keeps the plant upright.	Cannot become turgid.
Cannot move from place to place due to rigidity of the cell wall.	Lacking a cell wall, animal cells are flexible and can move to find shelter, food, or better reproductive opportunities.

0.6 marks per point (any 5) = 3 marks

Stem cells are undifferentiated, unspecialized cells that can give rise to other cell types. The **zygote** is the most basic stem cell — around 220 types of cells in the human body all develop from this single cell. **1 mark**

Haematopoietic stem cells produce red blood cells, white blood cells, platelets, muscle cells, nerve cells, and liver cells. **1 mark**

3(iii) Comparison table: any EIGHT differences between mitosis and meiosis. 5 marks

Mitosis	Meiosis
Occurs in somatic cells.	Occurs in germ line cells.
Cell divides only once.	Cell divides twice.
Produces two daughter cells.	Produces four daughter cells.
Daughter cells produced are diploid.	All daughter cells are haploid.
Daughter cells become part of the somatic body.	Daughter cells form gametes.
All chromosomes remain independent of each other.	Homologous chromosomes pair with each other (synapsis).
No chiasmata formation.	Chiasmata are formed.
No crossing over occurs during the process.	Crossing over takes place during meiosis I.
Daughter cells are genetically alike.	Daughter cells are genetically different from each other and from the parent cell.
In most cases daughter cells can divide again.	Gametes produced by meiosis cannot divide any more.

0.625 marks per correct row (any 8 = 5 marks)

OR

Spindle apparatus formation (role of centrioles; plant cells) and significance of mitosis in regeneration, asexual reproduction, and cloning/tissue culture. 5 marks

The **spindle apparatus** is formed by the cytoskeleton of the eukaryotic cell during cell division, mainly from microtubules. It separates chromatids/chromosomes and moves them to opposite poles, equally distributing hereditary material into the daughter cells. **1 mark**

A pair of **centrioles** near the nucleus of the animal cell duplicates, and both pairs migrate to opposite poles, giving rise to spindle fibres. The spindle forms around the nucleus, but as soon as the nuclear envelope disintegrates, the fibres attach to the centromere of each chromosome and pull it to its pole. In **plant cells**, there are no centrioles, so the spindle apparatus forms **without centrioles**. **1 mark**

Regeneration: some animals can regenerate a whole missing body part — e.g. a **sea star can regenerate an arm**, **earthworms can regenerate the head**, and a **salamander can regenerate its limb** — the production of new cells to form missing parts involves mitosis. **1 mark**

Asexual reproduction: mitosis is the basis of asexual reproduction, which involves only one parent and produces genetically identical offspring; **vegetative propagation** is common in plants (e.g. potato, onion, garlic, ginger), and animals like **sponges, planaria, and hydra** reproduce by budding. **1 mark**

Cloning and tissue culture: mitosis has made it possible for scientists to produce a very large number of identical copies of living organisms in an artificial environment — called cloning or tissue culture if a small mass of tissue cells is used. **1 mark**

3(iv) Structure of lipids (fatty acids + glycerol), saturated vs unsaturated fatty acids, and two functions of lipids. 5 marks

Lipids are mainly made up of **fatty acids and glycerol** (an alcohol), collectively called acylglycerides. The most common type of acylglyceride is the **tri-acyl glyceride**, which contains one glycerol and three fatty acids (Glycerol + 3 fatty acids → Tri-acyl glyceride). **1.5 marks**

Saturated fatty acids have all C-C single bonds and are solid at room temperature, e.g. cheese, butter, ghee. **1 mark**

Unsaturated fatty acids have double or triple C-C bonds and are liquid at room temperature, e.g. soybean oil, mustard oil, olive oil. **1 mark**

Function 1: Phospholipids and cholesterol are components of the plasma membrane structure. **0.75 marks**

Function 2: Lipids act as energy stores in fat cells, the liver, and the blood; one gram of lipid provides **9.1 kcal** of energy — double the energy provided by carbohydrates or proteins. **0.75 marks**

OR

Classify lipids into fats/oils, waxes, and steroids with one example each; describe the tests for lipids. 5 marks

Fats/oils (triacylglycerides): made of glycerol + 3 fatty acids; solid saturated examples like **butter, ghee**, liquid unsaturated examples like **soybean oil, olive oil**. **1 mark**

Waxes: obtained from the comb of insects, used by them to construct hives; also form the waterproof, waxy cuticle covering the surface of leaves. **1 mark**

Steroids: lipids with a characteristic three-ring structure and no fatty acids, e.g. **cholesterol, vitamin D**. **1 mark**

Spot test: a drop of the test (lipid) solution is placed on filter paper and allowed to dry — a clear or greasy/translucent spot confirms that lipid is present. **0.67 marks**

Water (melting) test: fat (e.g. butter/ghee) is heated with distilled water in a test tube — the fat melts but remains insoluble, forming a separate layer over the water, confirming that lipid does not mix with water even in melted form. **0.67 marks**

Emulsion test: the test material is mixed with absolute alcohol and shaken, then cold water is added and shaken again — a **cloudy white suspension (emulsion)** confirms the presence of lipid. **0.66 marks**